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THE EFFECTS OF 13 DAYS MATERNAL SEPARATION ON INFANT RHESUS MONKEYS COMPARED WITH THOSE OF SHORTER AND REPEATED SEPARATIONS

BY YVETTE SPENCER-BOOTH & ROBERT A. HINDE

Medical Research Council Unit on the Development and Integration of Behaviour, Madingley, Cambridge

Abstract. The effects of removing the mothers of rhesus monkey infants for 13 days were qualitatively similar to those found previously, when the mothers were removed for 6 days. The infants' behaviour changed markedly in the first few days of separation: thereafter there was no clear evidence, from the measures used, of any further changes. After the mothers' return these infants gave more distress calls, and showed a greater depression of locomotor and play behaviour, than did the infants whose mothers had been removed for only 6 days.

Removal of the mother for 6 days from 21 to 32 week-old rhesus monkeys has marked effects on their behaviour during separation and for at least several weeks afterwards (Spencer-Booth & Hinde 1967, 1971). Although similar effects have been found by other authors (e.g. Seay, Hansen & Harlow 1962; Seay & Harlow 1965) and for other species (e.g. Kaufman & Rosenblum 1967, 1969), there have so far been no systematic attempts to assess the influence of the length of the separation period. In this experiment rhesus mothers were separated from their infants for 13 days: the effects on infant behaviour and on mother–infant interaction during the mother's absence and the 4 weeks after her return are compared with those of one or two 6-day periods of separation, as reported previously.

Methods

The infants were reared in groups containing one adult male with two to four females and their young, kept in outdoor cages (5·50 × 2·44 × 2·44 m) connecting with indoor rooms (2·29 × 1·83 × 1·37 m: see also Hinde & Rowell 1962). The experimental groups consisted of six long separation (LS) infants (four female and two male) separated at 30 to 32 weeks for 13 days, ten twice separated infants (TS) (four female and six male) separated at 21 or 25 to 26 weeks for 6 days and again at 30 to 31 weeks for 6 days, and five short separation (SS) infants (three female and two male) separated at 30 to 32 weeks for 6 days. (Two of the TS infants collapsed during their first separation, and the mothers were returned prematurely). The data for the TS and SS infants have been described in detail elsewhere (Spencer-Booth & Hinde 1971) and are discussed here only in comparison with the data for the LS infants.

The separation experiments involved removing the mother from the group and placing her in an indoor room about 55 m from and out of sight of the group for 13 or 6 days. Although routine data on mother–infant interaction were collected earlier and later, we are concerned here only with the 3 to 4 days preceding the mother's removal, the separation period, and the four weeks after the mother's return. In the case of the TS infants, the only data used were those concerned with their second separation period. Thus after the separation considered here the LS infants had undergone 13 days continuous separation, the TS infants two periods of 6 days separation with a gap of several weeks between them, and the SS infants 6 days separation only. Table I shows the days on which the different groups were watched.

Pre-Separation Period

Each infant was watched on 3 of the 4 days before the mother was removed (days 1 to 3) starting at 10.00 hours (two of the SS infants were watched only on 2 days). Routine data on mother–infant interaction were collected, and the infant's activity while off its mother was recorded (see Spencer-Booth & Hinde 1967). The watch was continued until the infants had been off the mother and all other animals for a total of 2 hr, or until 14.30 hours, whichever was the earlier.

Separation Period

At 10.00 hours on day 4 the mother was removed. The infant was watched as on days

Table I. Days on Which Each of the Three Groups were Watched

	Pre-separation period			Separation period							Post-separation period							
LS	1	2	3	4	5	9	12	15	16		17	18	19	21	23	30	37	44
TS	1	2	3	←	→	4	5	7	9	↔	10	11	12	14	16	23	30	37
SS	1	2	3	4	5	7	9	←	→		10	11	12	14	16	23	30	37
				Mother out						Mother in								

LS=long separation infants. TS=twice separated infants (second 6-day separation). SS=short separation infants. Days linked by ←→ were consecutive. In each vertical column after 'Mother-out' the total previous separation experience is equal. The post-separation data were lumped into periods indicated by the discontinuous lines.

1 to 3. It was watched similarly on days 5, 9, 12, 15 and 16 (LS infants) or on days 5, 7 and 9 (TS and SS infants). (For analysis, this separation period was divided up in different ways according to the nature of the problem: these are specified as appropriate in the text.)

Post-Separation Period

This was considered in three parts:

Day of mother's return. At 10.00 hours on day 17 (LS infants) or day 10 (TS and SS infants) the mother was returned. Data collection began immediately as on days 1 to 3.

Second post-separation period. Similar data were collected on days 18, 19, 21 and 23 (LS infants) or days 11, 12, 14 and 16 (TS and SS infants).

Third post-separation period. Similar data were collected on days 30, 37 and 44 (LS infants) or days 23, 30 and 37 (TS and SS infants).

It will be noted that the first post-separation period is only 1 day in length, while the others were about 1 and 3 weeks respectively.

Several observers were involved in the separation experiments, all of whom were extensively trained beforehand. The categories of behaviour used were so clear-cut that inter-observer differences were negligible and an assessment of inter-observer reliability unnecessary.

Measures

Data involving twenty-six measures of mother–infant interaction or infant behaviour were analysed: these are described in Spencer-Booth & Hinde (1971). For brevity, data on some of these were omitted from this paper. Those omitted were either concerned with aspects of mother–infant interaction which showed no differences between groups, or were redundant here because highly correlated with measures which were used. The remainder fall into the following groups:

Vocalizations

Whoos. The mean number of whoo calls per hundred half minutes in which the infant was off its mother.

Squeak/gecker/screech (S/G/S). The percentage of half minutes in which the infant was off in which one of this assembly of intergrading noises was recorded.

Measures of the Amount and Type of Activity

Locomotor activity. The number of 'boxes' entered by the infant per half minute in which it was off its mother.

Activity when active (>5/>1 per cent). The number of half minutes in which the infant entered more than five boxes, as a percentage of the number in which it entered more than one. (This is the equivalent of 100 minus the measure 2 to 4/>1 used in our previous papers. It is preferable because the percentage is higher, the more active the infant.)

Sitting (per cent S). The number of half min in which the infant was recorded sitting still and not engaged in any obvious activity, as a percentage of the number in which it was off.

Manipulative play (per cent SF). The number of half minutes in which the infant was engaged in manipulative activity, such as sitting fiddling with pieces of wire or chain, as a percentage of the number it was off.

Rough-and-tumble social play (per cent R & T). The number of half minutes in which the infant engaged in rough-and-tumble social play, as a percentage of the number it was off.

Approach/Withdrawal social play (per cent A/W). The number of half minutes in which the

infant engaged in approach/withdrawal social play, as a percentage of the number it was off.

Measures of Infant's Relationships with Group Companions

Leaning against. The percentage of half minutes off in which the infant was in close physical contact with a group companion other than the mother.

Time on group companions. The percentage of half minutes watched in which the infant was recorded as clinging ventro-ventrally to a group companion other than the mother.

Measures of Infant Behaviour on Mother and Time Spent Off

Asleep. The percentage of half minutes watched in which the infant was on the mother with its eyes closed.

Time off. The percentage of half minutes watched in which the infant was off the mother and all other animals.

Measures Relating to the Relative Roles of Mother and Infant in Determining Time Off

Every time the infant got on or tried to get on the mother we recorded whether it was through the mother's initiative (M) or the infant's: in the latter case the infant could be accepted (A) or rejected (R). This provides two measures:

The Frequency of Rejections (R). The mean number of rejections by the mother per hundred half minutes watched.

The Relative Frequency of Rejections (R/(A+M+R)). The proportion of the total number of times the infant got on the mother, or attempted to do so, which were rejected.

Measures of Where Infant was When Off Mother

>60 cm. The percentage of half minutes in which the infant was off in which it was recorded more than 60 cm from the mother. (60 cm is roughly the distance a mother can reach in a hurry, and was assessed in relation to the 120 cm divisions of the wire netting.)

Leaning against mother. The percentage of half minutes off in which the infant was in close physical contact with the mother whilst off.

Measures of Relative Responsibility for Maintenance of Proximity

Approaches (%Ap.). The percentage of occasions that the distance between mother and infant decreased from more than 60 cm to less than 60 cm (approaches) which were due to movement by the infant.

Leavings (%L). The percentage of occasions that the distance between mother and infant increased from less than 60 cm to more than 60 cm (leavings) which were due to movement by the infant.

%Ap.–%L. The difference between the proportions of approaches and of leavings which were due to movement by the infant. This gives a measure of initiative in maintaining proximity which is relatively independent of the relative activity of the two animals (Hinde & Atkinson 1970).

Following (%F). The percentage of half minutes in which the infant was off its mother in which it was recorded as following the mother around the pen.

Method of Analysis

For each animal the data were analysed separately for each watch. Watches were then grouped in the periods described above (see vertical lines in Table I): where periods contained more than one watch, each individual's mean was calculated from its score on each watch. There were of course no data on mother–infant interaction during separation, and some TS and LS infants were off their mothers so little on the day she was returned that measures related to time off mother were unreliable.

The effects of the 13-day separations on the LS infants were assessed by comparisons of individual mean pre-separation scores with those in each separation and post-separation period. Changes during the separation period were assessed by comparisons of scores between days 5, 9 and 16. In both cases the significance of differences was assessed with the Wilcoxon matched-pairs signed ranks test (two-tailed).

For comparisons between groups, the Mann-Whitney *U* test (two-tailed) was used. Further divisions of the separation period of the LS infants used in these comparisons are mentioned later.

Results

Infant Mortality

Two TS infants died before they were a year old: one at 50 and one at 47 weeks (post mortem findings described in Spencer-Booth & Hinde, 1971). One LS infant died at 41 weeks: there were moderate accumulations of chronic inflammatory cells around the larger lung vessels, with mild lymphoid hyperplasia, and pulmonary

oedema. It was thought that the severity of these lung changes would not normally be sufficient to cause death, and no definite cause of death was found.

A Case of Delayed Re-Establishment of Mother-Infant Relations

It is necessary to consider first one of the LS females (Daphne) whose post-separation data were qualitatively different from the others. The difference lay in the very close relationship she established with a nulliparous female in the group while her mother was out, and in the delayed re-establishment of her relationship with her mother.

On the last two days of separation (days 15 and 16) she spent more time on the female group companion (87 per cent) than she had spent on the mother before separation (74 per cent). When the mother was returned the infant came off the female and ran to the mother as she came into the pen. However the mother was viciously attacked by the female who had been carrying the infant, and the rest of the group joined in. A little later the infant was rejected by her mother when she tried to get on her, and when she did get on was put down by the mother as soon as the others came near. The infant then began going on other group companions and later still was seen to run right past the mother, ignoring her. Time off group companions was 57 per cent and time on mother 3 per cent on this day.

The following day (18) the group had settled, with the mother and infant seeming quite independent; the infant was never on the mother during the watch and the relationship with the other female seemed to be continued unabated. This situation seemed to result from a combination of this female's possessiveness of the infant and the mother's rejection of her. On day 19 the infant was again never on its mother during the watch, although she made some attempts to go to her which were prevented by the other female. The infant licked the mother's wounds and followed her, but the mother either did nothing or avoided the infant.

On days 21 and 23 the situation remained similar: the infant was hardly on the mother at all, but was on the other female a great deal. The infant looked unwell.

However by the next week, and for the remaining watches of the experiment, the infant was re-established with its mother (on her for 81, 83 and 95 per cent of watch) and was on the other female very little. The infant's health remained poor through this period.

The unusual fighting and injury to the mother undoubtedly played a large part in the delayed re-establishment of her relationship with the infant: rejection seemed to be largely by the mother and not by the infant. In a number of measures the data for this infant were considerably different from that of the other five infants separated for 13 days. These cases are noted in the discussion of the effects of separation on these infants (see below).

Comparison of Separation and Post-Separation with Pre-Separation Data for LS Infants

We are concerned here with the changes from the pre-separation period produced by the separation experience in the LS infants. Since the changes were qualitatively similar to those reported previously for 6-day (TS and SS) separations (Spencer-Booth & Hinde 1971), all the data are not given in detail though many are to be found in Table II: quantitative comparisons between the groups are given later. In those cases in which all members of the LS group showed changes in the same direction, the difference is significant ($P<0{\cdot}05$, two-tailed) for the group as a whole.

Vocalizations. Whoo calling was more frequent than before separation on the day the mother was removed and over the separation period as a whole (i.e. including day 4) for all infants. The whoo-calling rate remained above the pre-separation level on the day the mother was returned (day 17) and in the second post-separation period for all infants except Daphne.

Measures of activity. Over the separation period as a whole locomotor activity was less than before separation in all animals (No. of boxes less, $>5/>1\%$ lower, % sitting higher). The change in % sitting was apparent on day 4, although the change in the other two measures was not. All the LS infants ate more and played less during separation, although only social played decreased on day 4.

On the day the mother was returned all these changes were still present except for that in sitting. Number of boxes and social play remained below pre-separation levels in all post-separation periods: the change in manipulative play was still present only in the second and that in $>5/>1\%$ in the third post-separation period.

Measures of relations with group companions. Both the time spent on group companions by

Table II. Changes Within the 13-day Separation Period

	Day 4 median	Day 9 median	Day 9 versus day 4 ±	Day 9 versus day 4 *N*	Day 16 median	Day 16 versus day 4 ±	Day 16 versus day 4 *N*	Day 16 versus day 9 ±	Day 16 versus day 9 *N*
Vocalizations									
Whoos/100 ½ min whilst off	217	37	−	6/6	40	−	6/6	+	4/6
S/G/S off	5·0	2·7	−	4/5(1=0)	10·1	+	5/6	+	4/6
Amount and type of activity									
No. boxes/½ min	2·5	1·3	−	6/6	1·4	−	6/6	+	4/6
>5/>1%	18	2	+	5/6	5	+	5/6	+	3/6
% sitting	77	60	−	5/6	60	−	3/6(1=0)	·	3/6
% eating	30	40	+	4/6	35	−	5/6	−	3/6
% sitting fiddling	8	2	−	5/6	0·2	−	5/5(1=0)	−	3/6
% R&T play	0	0	−	2/2(4=0)	0	−	2/2(4=0)	−	1/1(5=0)
% A/W play	0	0	−	2/2(4=0)	0	−	2/2(4=0)	−	1/1(5=0)
Group companions									
Time leaning against	7·5	27	+	6/6	35	+	4/6	+	3/6
Time on	0	0	+	2/2(4=0)	9	+	4/4(2=0)	+	4/4(2=0)

Figures represent median or proportion changing in direction indicated by sign. 6/6 is significant with $P<0{\cdot}05$, Wilcoxon matched pairs test, two-tailed. Where *N* is less than six the number of animals whose score was 0 on the 2 days is indicated in parentheses.

the five infants which went on at all and the time spent leaning against them increased during separation but, with the exception of Daphne, the effect largely disappeared as soon as the mother was returned.

Measures of mother–infant relations

(a) Proportion of time on or off mother. On the day the mother returned and in the second post-separation period all animals except Daphne spent less time off than before separation, and more time asleep. By the third post-separation period these effects had disappeared.

(b) Measures relating to responsibility for time on or off. There was a tendency for the absolute frequency of rejections to be reduced on the day the mother returned and in the third post-separation period, but there were no significant differences with either this measure or the relative frequency of rejections.

(c) Proximity when off mother. There was a tendency for the infants to be at a distance from their mothers less, and to remain at a distance from her for fewer whole half minutes after separation, but this reached significance only for >60 cm in the second post-separation period. All animals except Daphne went >60 cm from their mothers less on the day she returned and spent fewer whole half minutes at a distance from her in the second post-separation period than before separation.

(d) Responsibility for proximity. After separation the lessened activity of the infants was reflected in their lower %Ap. and %L scores, particularly on the day the mother returned and in the second post-separation period. Although the %Ap.–%L median scores were higher after separation, the differences did not reach significance.

Changes Within the Separation Period

The SS and TS infants showed little evidence of any recovery during the 6-day separation except for a decrease in the rate of whoo-calling and amount of sitting. The present data are here examined to see whether any recovery could be detected in the longer period of separation used in this experiment. The data are given

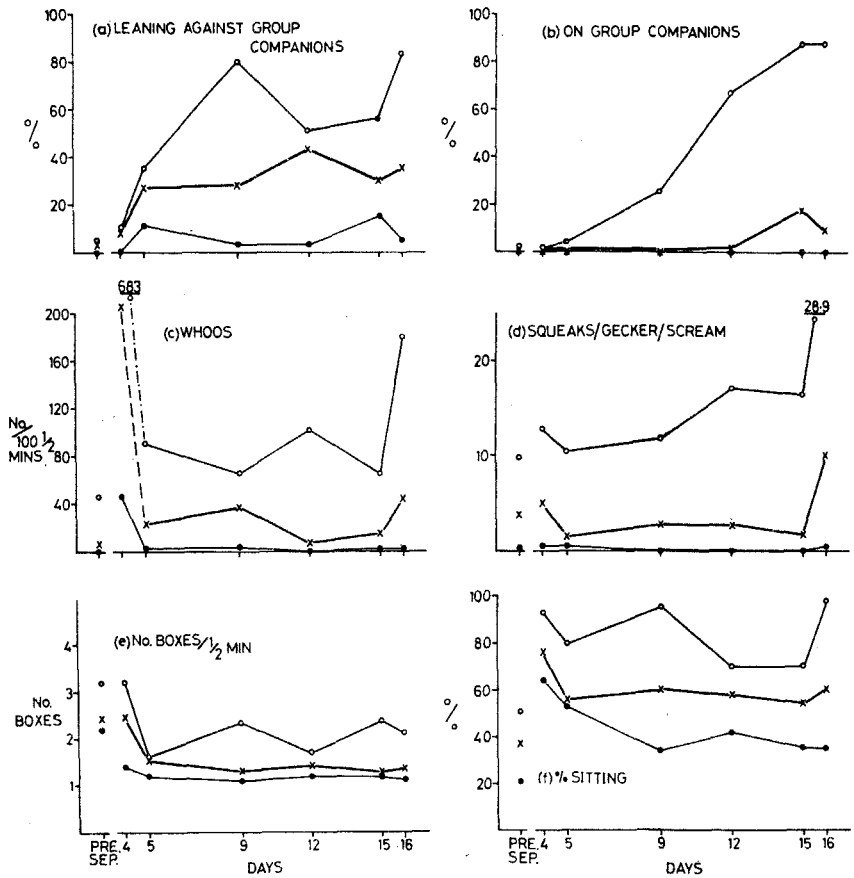


Fig. 1. Changes in the behaviour of long separated infants during the separation period. ×, Median for all LS infants; ○ and ●, individuals with extreme pre-separation values.

in detail in Fig. 1; comparison between the first day of separation (day 4), day 9 (=day 9 of 6-day separations) and day 16 are summarized in Table II.

The changes which occurred in the 6-day separation of the SS and TS infants were found also in the first 6 days of the longer separation of the LS infants. Thus the frequency of whoo-calls decreased from day 4 to day 9, and a similar trend was shown by squeak/gecker/screeches. Locomotor activity decreased from day 4 to day 9, though the change in activity when active (>5/>1%) did not reach significance. Sitting, however, tended to decrease, though not significantly. Social and manipulative play tended to decrease, though again the changes were not significant. The time leaning against group companions was higher on day 9 than on day 4, and two animals went on group companions on day 9, as compared with none on day 4. Thus although the distress calling produced by the mother's removal abated, the depression of locomotor activities increased over the first 6 days of separation.

Comparison of day 16 with day 9 revealed little evidence of recovery from this condition. While none of the measures showed a significant difference, the median rates of whoo-calls and S/G/S were higher on day 16 than on day 9. This increase occurred in spite of the fact that all animals which went on group companions at all went on more on day 16 than on day 9. The various measures of locomotor activity and of manipulative play showed no major changes between day 9 and day 16, though the only animal to show social play on day 9 did play less on day 16. It is thus clear that, over the group as a whole, the major changes in these measures

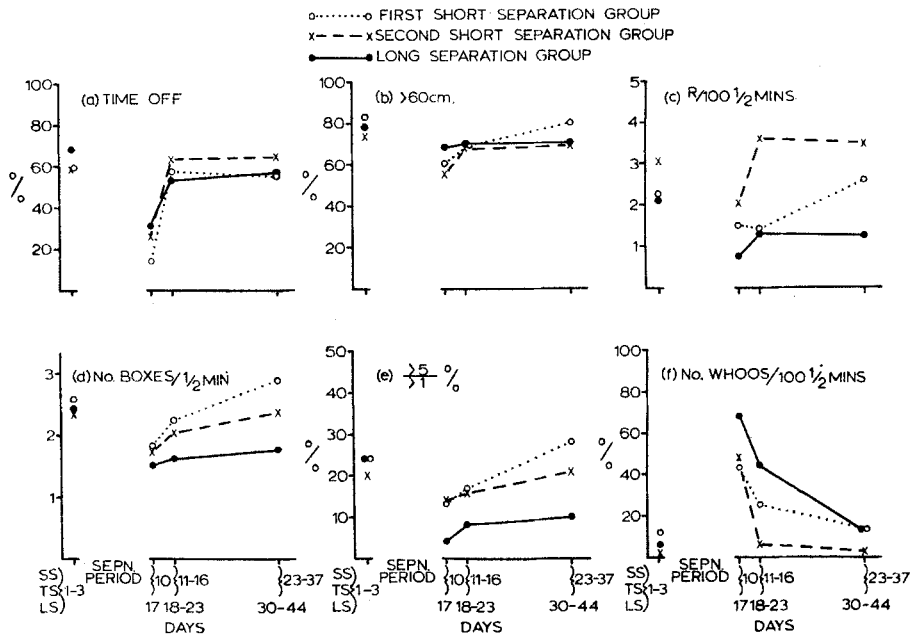


Fig. 2. Comparison between the pre- and post-separation data of the two short separation groups and the long separation group. The data are lumped as indicated in Table I.

occurred during the first 6 days after the other's removal, and that during the next 7 days there was neither substantial further deterioration nor recovery.

Comparison of the Effects of a Single 13-Day Separation with Those of One or Two 6-Day Separations

The data for the six LS infants, separated for 13 days at 30 to 32 weeks, are here compared with those for the second separation (6 days at 30 to 32 weeks) of the TS infants and the only separation (6 days at 30 to 32 weeks) of the SS infants (see Table III: some selected measures are shown in Fig. 2).

Pre-separation period. We may first compare the pre-separation scores of the three groups. Between the LS and the TS groups there were two significant differences from amongst the twenty-six measures examined: the LS infants were off their mothers more and spent less time asleep than the TS infants: these differences were perhaps consequences of the previous separation experience of the TS group. In any case, none of the measures of infant behaviour when off the mother showed significant differences between the two groups.

Between the LS and SS groups there was only one significant difference amongst the twenty-six measures: the LS group was recorded sitting more often ($P=0{\cdot}03$). Thus these groups were effectively similar before the mothers were removed.

Separation period. Turning now to the separation period itself, eight measures of infant behaviour are available. The means of days 9, 12, 15 and 16 of the LS infants were compared with those of days 4, 5, 7 and 9 of the TS infants: in each case there had been 6 days of separation from the mother beforehand, but with the TS infants a period with the mother had intervened (Table I). The TS infants were more active, more active when active, and were recorded sitting less, than the LS infants. They also showed more manipulative and (only just significantly with $P<0{\cdot}05$) more rough-and-tumble play. Overall there is thus clear evidence that these effects of the mother's absence were more marked in the second half of the continuous 13-day period than in the second of two 6-day separation periods.

With LS and SS infants, two comparisons were made. First, the means of days 5 and 9 were compared. At this point, both groups had had

Table III. Comparison of Medians of LS, TS and SS Infants

	Pre-separation			Separation					Post-separation								
									Period 1			Period 2			Period 3		
	LS	TS	SS	LS 9–16	TS 4–9	LS 5&9	LS 12–16	SS 5&9	LS	TS	SS	LS	TS	SS	LS	TS	SS
Vocalizations																	
Whoos	6	1	11	26	45	30	27	68	68	47	43	44	6*	25	13	2*	13
Squeak/Gecker/Screech	3·0	1·9	2·1	5·1	0·4	4·9	7·5	2·7	18·3	13·9	4·5	12·7	2·5†	6·1	9·3	2·2†	2·0
Amount and type of activity																	
No. boxes/½ min	2·4	2·4	2·6	1·3	2·0†	1·5	1·4	1·7	1·5	1·7	1·8	1·7	2·0†	2·3*	1·8	2·4†	3·2†
>5/>1%	23	19	23	5	13†	3	6	*13	3	14*	13	8	16	17	10	22†	28†
% *sitting*	37	23	16*	57	34†	63	58	*42*	63	39	12	50	29†	19†	49	22*	20†
% manipulative play	24	20	19	1	7†	3	0·2	6	2	2	4	5	15*	13	14	19	14
% R&T social play	3	6	13	0	0·6*	0	0	1†	0	0	2	0	3†	5†	0	5†	11†
% A/W social play	3	2	1	0·1	0	0	0	0	0	0	0	0	0·7†	1·1	0	1·0†	3·4†
Time on and off mother																	
Asleep	16	21*	15						42	43	25	21	21	20	23	19	16
Total time off	69	59*	59						31	26	14	53	64	57	56	65	55
Responsibility for time off																	
R/100 min	2·1	3·1	2·3						0·7	2·0	1·5	1·3	3·6†	1·4	1·3	3·5	2·6
Distance when off																	
>60 cm	79	75	83						69	55	60	70	68	70	71	70	80
Responsibility for proximity																	
% Ap. – % L	24	21	9						30	40	23	50	31	12	33	26	21
% F	4·4	4·7	1·1						1·3	5·5		4·1	6·1	2·2	3·1	5·6	3·9

Figures indicate group medians for period specified. * and † indicate difference from LS significant with $P<0{\cdot}05$ and 0·02 respectively (Mann–Whitney U Test, two-tailed). With SS data for separation period, symbol preceding median indicates significance of difference from LS 5&9, symbol following that of difference from LS 12–16.

the same treatment. In spite of that, the LS group were less active when active and sat more than the SS group: the difference in overall activity was in the same direction but not significant. The pre-separation differences in these measures were in the same direction, though only the difference in sitting was significant. The percentage changes from pre-separation to days 5 and 9 did not differ significantly. There is thus no clear evidence that these two groups responded differently to the mother's removal.

Second, the means of days 5 and 9 of the SS infants were compared with those of days 12, 15 and 16 of the LS group: here we are comparing behaviour immediately before the mother's return. The differences were in the same direction as those seen in the first comparison, but only that in time spent sitting was significant.

Post-separation period. (a) Vocalizations. After separation the LS infants whoo-called more than either the TS or the SS infants. With the TS infants the difference was not significant on the day the mother was returned, but was in each of the subsequent periods. With the SS infants, none of the differences were significant.

S/G/S were consistently higher for the LS infants than for the TS infants, the differences becoming significant in the latter part of the post-separation period. That the latter was not merely a consequence of the slight (non-significant) pre-separation difference is suggested by the greater percentage change shown by LS infants for the two last post-separation periods. Post-separation differences between LS and SS groups were in the same direction, but did not reach significance.

Tantrums were for the most part too infrequent for comparisons to be made, though the LS infants gave significantly fewer on the day the mother was returned than did the TS infants.

(b) Measures of activity. During the post-separation period the LS infants were less active and less active when active, than either of the other groups, the differences being significant in four out of six cases for TS and three out of six for SS.

The data for the percentage of half minutes in which the infants were recorded sitting yielded a similar picture. Although there had been no difference between LS and TS infants before separation, after separation the LS infants sat significantly more in two out of three periods. The LS infants sat more than the SS infants before separation, as well as in all succeeding periods, so here the post-separation difference cannot certainly be ascribed to the separation experience: the two groups showed no significant differences in percentage change.

Thus the LS infants became more depressed in activity during separation than either the TS or SS infants, and remained less active during the post-separation periods.

Manipulative play in the LS group during the latter part of its separation period and in one of the subsequent periods was significantly lower than that of the TS group. The differences from the SS group were qualitatively similar, but not significant.

Social play was too infrequent in some periods for comparisons to be possible. However there were no significant differences between the LS and other groups before separation, but LS was significantly lower in the majority of the subsequent periods for which comparisons were possible.

(c) Proportion of time off and at a distance from mother. There were no significant differences between the LS and TS infants in time off the mother or time at a distance from the mother. The small differences present before separation had thus disappeared. There were no differences in any of these measures between LS and SS infants either.

(d) Responsibility for time off. The LS infants were rejected less frequently than either the TS or SS infants at all stages in the experiment, though the only significant difference was for the LS versus TS comparison in the second post-separation period. That this was not merely a reflection of the pre-separation difference is suggested by the finding of a significant difference in the percentage change in absolute frequency of rejections from the pre-separation period to both this period ($P<0{\cdot}05$) and to the subsequent one ($P<0{\cdot}02$). The LS infants also showed a greater percentage change in both absolute and relative frequency of rejections than the SS infants for the third post-separation period ($P=0{\cdot}03$ and $=0{\cdot}05$, respectively).

(e) Responsibility for proximity, Throughout the post-separation period the LS infants tended to have lower %Ap. and %L scores than either the SS or the TS infants, the differences being significant in nearly all periods. This, however, was probably solely a reflection of the greater reduction in their locomotor

activity (see below): at no stage did the LS infants differ significantly in %Ap.–L% from the other groups. There were no differences between the groups in percentage of half minutes in which they followed the mother (%F) or in the time spent leaning against her (data available for LS and TS only). However on the day the mother returned the LS infants had a lower median %F score than before separation and the TS and SS infants a higher one (percentage change significantly different between LS and TS ($P=0{\cdot}02$)). The same was true of time spent leaning against the mother on the day she returned (percentage change again significantly different, $P<0{\cdot}05$). This, coupled with their smaller (but not significantly so) %Ap.–%L score, suggests that on the day of the mother's return the LS infants played a smaller relative role in maintaining proximity to their mothers than did the TS infants. In the other two post-separation periods, however, most of these differences were reversed and the LS infants tended to play the greater role in maintaining proximity.

Differences Between Groups: Summary

There were thus no significant differences between LS and either TS or SS infants in the times the infants spent on or near the mother. There were some indications that the LS infants were rejected less and, after the day of the mother's return, played a greater part in maintaining proximity than did the infants of the other two groups.

While there were no dramatic differences between groups in mother–infant relations, there were in the behaviour of the infants when off their mothers. Measures of the amount and type of activity indicated that the effects of the long separation were considerably more severe than those of either one or two short separations, even though in the latter case the total period of separation was about the same.

Discussion

The effects of removing the mother for 13 days were qualitatively similar to those found in a previous study in which they were removed for only 6 days. Within the separation period itself there was no clear evidence, from the measures which we used, that the changes seen over the first 6 days either continued or were reversed during the rest of the period. Rather, after the first days the calling and other types of activity studied seemed to reach a plateau and to change little during the rest of the 13 days. There was no substantial evidence of a recovery comparable to that which Kaufman & Rosenblum (1967) reported to begin in pigtail macaques after 5 to 6 days separation.

Does this mean that the effects of the separation experience are much the same for 6- and 13-day separations? Comparison of the post-separation data between infants separated from their mothers for one or two periods of 6 days with those separated for 13 days showed only minor differences in the post-separation mother–infant relations, though the smaller frequency with which the LS mothers rejected their infants after separation suggest that they were more amenable to their infants' demands than were the mothers of the other groups. When the behaviour of the infants themselves is examined, however, there are clear indications that the infants separated for one period of 13 days were more severely affected than those separated for one or two periods of 6 days. Their distress calling remained higher for at least a week after their mother was returned, while their locomotor activity and probably their play behaviour was more depressed for at least a month. Indeed while the infants which had had one 6-day separation were showing at least as much locomotor activity a month after the mother's return as they had done before separation, those separated for 13 days were showing significantly less. The infants separated for two periods of 6 days occupied an intermediate position on this measure.

Thus these data not only confirm the previous finding that a period of separation from the mother can produce depression of a number of types of activity, but indicate that the severity and durability of the depression are related to the length of the separation period.

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REFERENCES

Hinde, R. A. & Atkinson, S. (1970). Assessing the roles of social partners in maintaining mutual proximity, as exemplified by mother–infant relations in rhesus monkeys. *Anim. Behav.*, **18**, 169–176.

Hinde, R. A. & Powell, T. E. (1962). Communication by postures and facial expressions in the rhesus monkey (*Macaca mulatta*). *Proc. zool. Soc. Lond.*, **138,** 1–21.

Kaufman I. C. & Rosenblum, L. A. (1967). Depression in infant monkeys separated from their mothers. *Science, N.Y.*, **155,** 1030–1031.

Kaufman, I. C. & Rosenblum, L. A. (1969). Effects of separation from mother on the emotional behaviour of infant monkeys. *Ann. N.Y. Acad. Sci.*, **159,** 681–695.

Seay, B. & Harlow, H. F. (1965). Maternal separation in the rhesus monkey. *J. nerv. ment. Dis.*, **140,** 434–441.

Seay, B., Hansen, E. & Harlow, H. F. (1962). Mother–infant separation in monkeys. *J. Child Psychol. Psychiat.*, **3,** 123–132.

Spencer-Booth, Y. & Hinde, R. A. (1967). The effects of separating rhesus monkey infants from their mothers for six days. *J. Child Psychol. Psychiat.*, **1,** 179–197.

Spencer-Booth, Y. & Hinde, R. A. (1971). Effects of six-days separation from mother on 18-32 week-old rhesus monkeys. *Anim. Behav.*, **19,** 174–191.

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